

Unit 5 Mini-Cases

Principles of Capital Markets

Winter Term 2025/26

Please upload your solutions and indicate which Mini-Cases (MCs) you have solved before **November 19, 2025, at 6am**. To do so, go to the assignment section on Canvas and choose “Unit 5 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions upload will be considered, your last complete attempt counts!** You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. one pdf and one Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Important remark for the presentations: Make sure to provide a concise explanation of your solutions. In particular, when applying any formula from the lecture materials or other sources, you may be asked to explain the structure of your solutions and provide the intuition and underlying concepts behind the formula used.

Mini-Case 5.1 (pdf or Excel or R)

- (a) Calculate the Bloomberg-adjusted betas for the following companies. Explain the reasoning behind such a beta adjustment. For which company is the absolute adjustment effect the largest/smallest?

Company	Unadjusted beta
OMV	0.49
Bayerische Motoren Werke AG	0.77
Goldman Sachs Group Inc.	1.35
Palantir Technologies Inc.	1.48
Trump Media & Technology Group	4.61

- (b) Think of one company for which you expect a low beta and one company for which you expect a high beta. Then, go to Yahoo Finance ($\rightarrow$ “Statistics”) and report the companies’

respective beta estimates (over 5 years of monthly data) and calculate the corresponding Bloomberg-adjusted betas. Do the reported values surprise you?

Mini-Case 5.2 (pdf)

- (a) Identify each of the following risks as most likely to be systematic risk or diversifiable risk:
- (i) The risk of a rare earths shortage.
 - (ii) The risk of cyber attacks.
 - (iii) The reputation risk from selecting a politically outspoken board member.
 - (iv) Stagflation risk.
 - (v) The risk of labor disputes.
- (b) Give an example of financial risk that entails both a systematic and a diversifiable component.

Mini-Case 5.3 (pdf)

Consider the following financial information about the risk-free rate, three stocks, and the market portfolio:

	Expected return	Standard deviation	Correlation with market	Beta
Stock A	(1)	(i)	0.40	0.80
Stock B	(2)	0.25	(ii)	1
Stock C	(3)	0.50	0.50	(iii)
Market portfolio	0.08	0.20	(iv)	(v)
Risk-free asset	0.01	(vi)	(vii)	(viii)

- (a) Fill in the missing numbers (i)-(viii) as implied by the CAPM.
- (b) Now, assume the following expected returns: (1) = 6%, (2) = 8%, and (3) = 11%. Are the three stocks correctly priced? If not, what price adjustments do you expect going forward?

Mini-Case 5.4 (pdf)

Consider the following two-factor model for the returns of two securities:

$$\tilde{r}_A = 0.02 + 0.5\tilde{F}_1 + 1.5\tilde{F}_2 + \tilde{\epsilon}_A$$

$$\tilde{r}_B = 0.02 + 1\tilde{F}_1 + 0.5\tilde{F}_2 + \tilde{\epsilon}_B$$

The variances of the two factors are as follows:

$$\text{var}(\tilde{F}_1) = 0.03, \quad \text{var}(\tilde{F}_2) = 0.04.$$

Assume that all two factors are independent (uncorrelated) and that all epsilons ($\tilde{\epsilon}$) have means of zero.

- (a) If $\text{var}(\tilde{\epsilon}_A) = \text{var}(\tilde{\epsilon}_B) = 0.05$, what are the return variances of the two securities, as well as the covariance and correlation between them?
- (b) How much should you invest in each of the above two securities to design a portfolio with a beta on $\tilde{F}_2$ twice as large as on $\tilde{F}_1$? *Hint: The portfolio weights should sum to one. For an example, see Example 6.5 in the textbook.*

Mini-Case 5.5 (pdf)

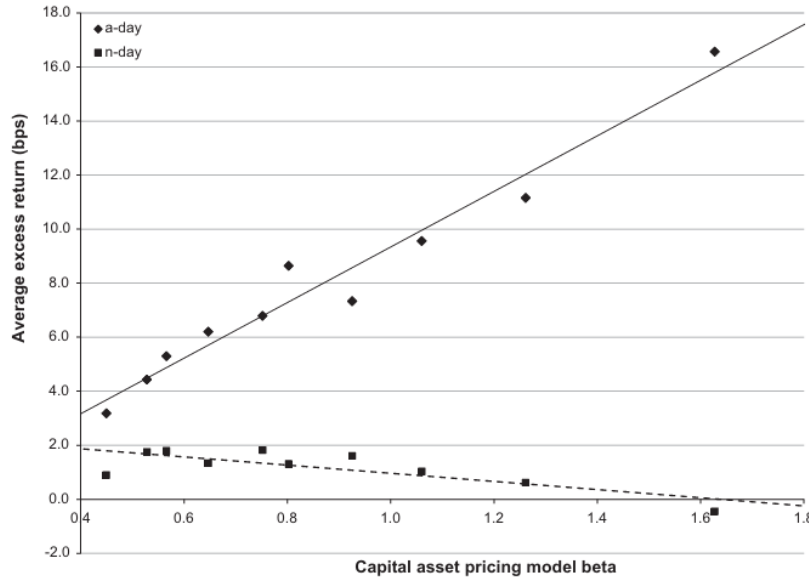


Fig. 1. Average excess returns for ten beta-sorted portfolios. This figure plots average daily excess returns in basis points (bps) against market betas for ten beta-sorted portfolios of all NYSE, Amex, and Nasdaq stocks separately for announcement days or a-days (days on which inflation, employment, or Federal Open Market Committee interest rate decisions are scheduled to be announced) and non-announcement days or n-days (all other days). The implied ordinary least squares estimates of the securities market line for each type of day are also plotted. The sample covers the 1964–2011 period. For each test portfolio, the same estimate of its full-sample beta is used for both types of day.

The figure above is from an influential paper by Savor and Wilson (2014).¹ The authors analyze 10 stock portfolios with different average beta values, ranked from low to high. For these portfolios they then compare their average beta values to their average excess returns for announcement days (“a-days,” see caption for details) and non-announcement days (“n-days”). What does the above figure tell us about the empirical performance of the CAPM?

Mini-Case 5.6 (Excel or R)

In their seminal work, Eugene Fama and Kenneth French demonstrated that, historically, both

1. Small cap stocks and
2. Stocks with high book-to-market ratios

¹Savor, P., & Wilson, M. (2014). Asset pricing: A tale of two days. *Journal of Financial Economics*, 113, 171-201.

have outperformed the market. They argued that in order to explain an asset's return, one needs to account for its exposure relative to a portfolio of “small (market cap) minus big” stocks (SMB) and “high (book-to-market) minus low” stocks (HML). This has led to the so-called “Fama-French three-factor model:”

$$\tilde{r}_{it} = \alpha_i + \beta_i(\tilde{R}_{Mt} - r_f) + \delta_i \widetilde{\text{SMB}}_t + \gamma_i \widetilde{\text{HML}}_t + \tilde{\epsilon}_{it},$$

where $\widetilde{\text{SMB}}_t$ denotes the performance of the SMB portfolio in period t and $\widetilde{\text{HML}}_t$ the performance of the HML portfolio in period t .

In the spreadsheet “MarketDataUnit6.csv” you can find:

- (i) Monthly stock returns for Costco (COST), IBM (IBM), and Southwest Airlines (LUV) from January 2001 to December 2024
- (ii) Monthly returns of the market portfolio (over the risk-free rate) as well as monthly returns of the SMB and HML portfolios from January 2001 to December 2024²

Based on this data, perform the following analysis:

- (a) Estimate the parameters α_i , β_i , δ_i , and γ_i of the Fama-French three-factor model for all three stocks. *Hint: To estimate a linear regression in Excel, go to Data Analysis > Regression.*
- (b) Assume the current risk-free rate is 1% per annum. Based on your estimates in (a), compute the expected monthly and annual return for each stock. *Hint: Use the average value of each factor to estimate the different risk premia.*
- (c) In your opinion, how well does the Fama-French three-factor model perform? Does it provide significant explanatory power in addition to the CAPM?

²This data is publicly available from Kenneth French's website <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/index.html>.